Donut iso variants: inner-R truth-purity scan and ABCD closure

All-calo 120 MeV donut with $R_{\mathrm{outer}} = 0.4$ (cross-check)

PPG12 Analysis Team (automated)

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Executive summary

Two new reco-isolation definitions were added as cross-check variants (`syst_type=None`, `syst_role=None`):

- **donutFull** (`use_topo_iso=4`): `cluster_iso_04` − `cluster_iso_{005,0075,02}`. All-calo 120 MeV tower sum (EMCal+HCalIn+HCalOut), outer $R = 0.4$, inner $R \in \{0.05, 0.075, 0.20\}$. Cluster ET is subtracted as a scalar in both terms and cancels in the difference.

- **donutExcl** (`use_topo_iso=5`): `cluster_iso_excl_04` − `cluster_iso_excl_{005,0075,01,02}`. Same three-calo 120 MeV sum, but EMCal towers owned by the cluster are skipped at tower granularity rather than subtracted as a scalar. Inner $R \in \{0.05, 0.075, 0.10, 0.20\}$.

Switch added in `RecoEffCalculator_TTreeReader.C:2042-2065`, mirrored in five companion macros. `donutFull_01` is deliberately skipped because `cluster_iso_01` does not exist in the slimtree (logged in `reports/anatreemaker_reprocess_TODO.md`, along with the EMCal-only inner-ring branches needed for a pure-EMCal donut).

Key finding: `donutFull_02` and `donutExcl_02` give $\sim 2.5\times$ better ABCD-on-MC closure than nominal topo04 iso (mean closure deviation $\langle |1 - r| \rangle \approx 0.07$ vs. nominal 0.185). Low-pT closure is worst across *all* variants (ratio 1.35–1.80 at 9 GeV), high-pT closure is 0.83–0.87 at 34 GeV.

Two-pass workflow

Per-variant iso cut was derived at 80 % target efficiency:

1. **Pass 1**: run the full `RecoEffCalculator_TTreeReader + MergeSim + CalculatePhotonYield` with placeholder loose cut ($b = 100, s = 0$).

2. **Cut derivation**: parameterized `FindETCut.C(var_type, target_eff)` fits $E_T^{\mathrm{iso,cut}} = b + s\,p_T$ at 80 %; `derive_iso_cuts.py` drives all 7 variants.

3. **Pass 2**: regenerate configs with derived (b, s), re-run pipeline on HTCondor cluster 2701654.

Table 1: Derived 80 % iso cuts (isoET $< b + s\,p_T$).

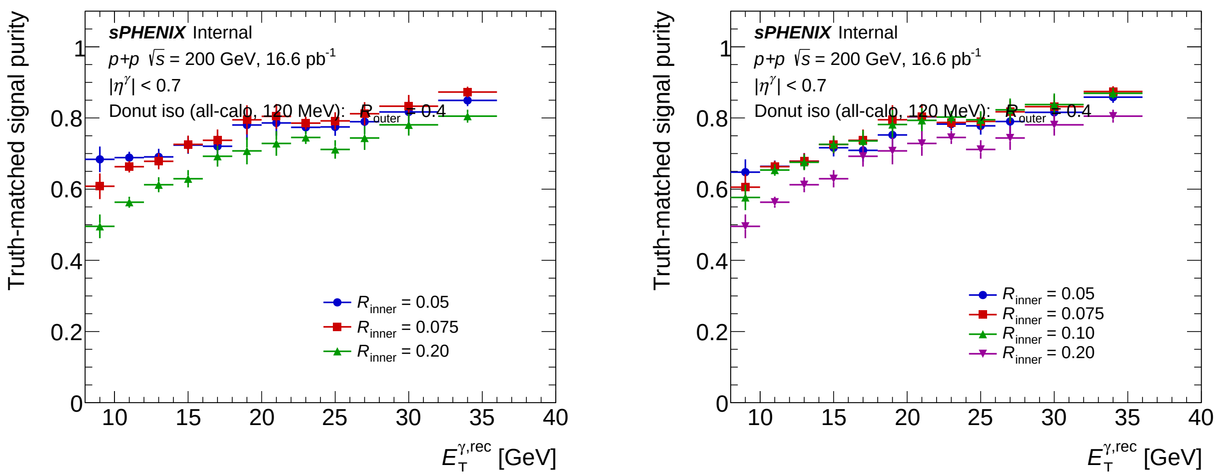
Variant	b [GeV]	s
donutFull_005	0.1089	0.0224
donutFull_0075	0.2642	0.0061
donutFull_02	0.2305	−0.0012
donutExcl_005	0.2457	0.0111
donutExcl_0075	0.2774	0.0053
donutExcl_01	0.2921	0.0017
donutExcl_02	0.2305	−0.0012

Truth-matched signal purity

Purity read from Photon_final_bdt_<var>_mc.root (MC-as-data, truth labels populated).

Table 2: Truth-matched photon purity at three representative p_T bins. Nominal topo04 iso included for reference.

Variant	$y(9\,\mathrm{GeV})$	$y(21\,\mathrm{GeV})$	$y(34\,\mathrm{GeV})$
nom (reference)	0.581	0.743	0.797
donutFull_005	0.684	0.786	0.849
donutFull_0075	0.608	0.805	0.873
donutFull_02	0.496	0.728	0.805
donutExcl_005	0.648	0.803	0.858
donutExcl_0075	0.606	0.804	0.874
donutExcl_01	0.577	0.793	0.869
donutExcl_02	0.496	0.728	0.805



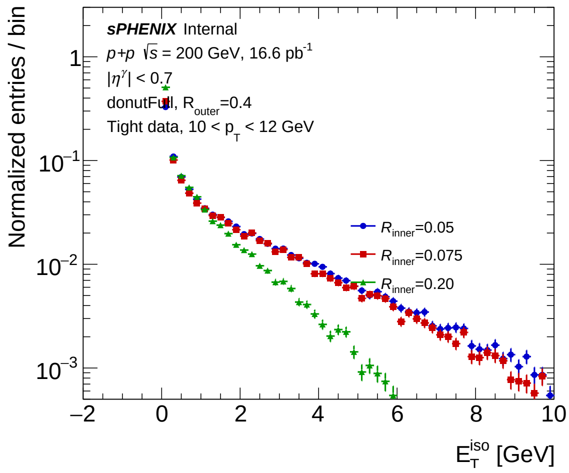
(a) donutFull family.

(b) donutExcl family.

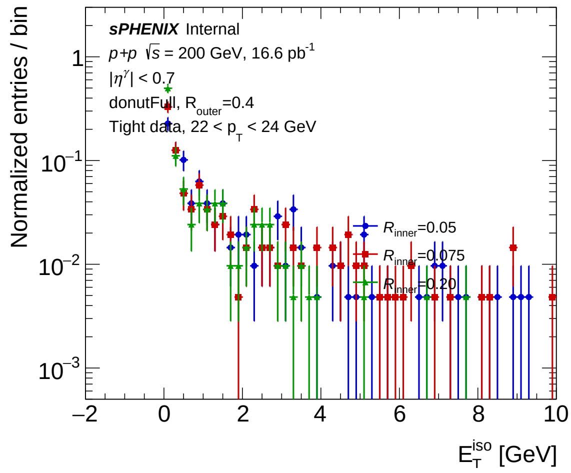
Figure 1: Truth-matched signal purity vs. cluster p_T for the donut-iso variants. Tighter inner R (smaller value) yields higher purity at low p_T; variants converge at high p_T.

isoET distribution by inner R (representative p_T bins)

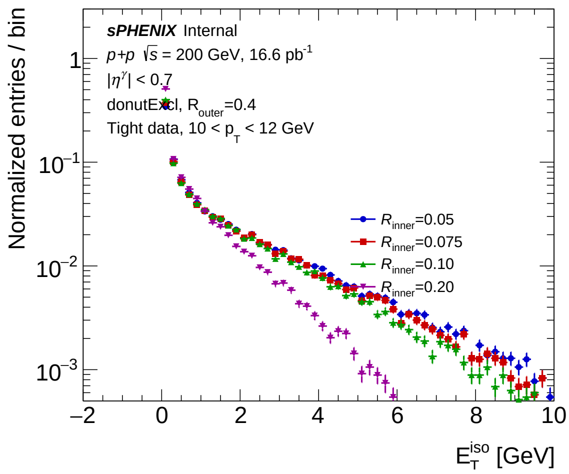
Overlay of the tight-shower-shape isoET distribution (h_tight_isoET_0_<ipt>, from data_histo_bdt_<var>.ro normalized to unit area, across donut inner R values at one low-p_T bin ($10 < p_T < 12\,\mathrm{GeV}$) and one higher-$p_T$ bin ($22 < p_T < 24\,\mathrm{GeV}$). Tight-region data is background-dominated below the iso cut; the distribution shape reflects the underlying-event pedestal of the donut definition.



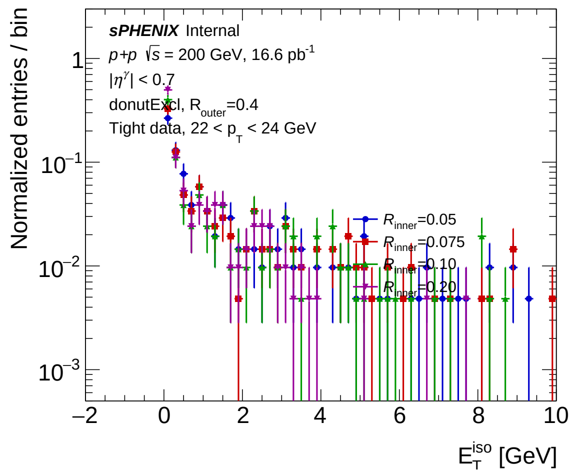
(a) donutFull, low p_T.



(b) donutFull, high p_T.



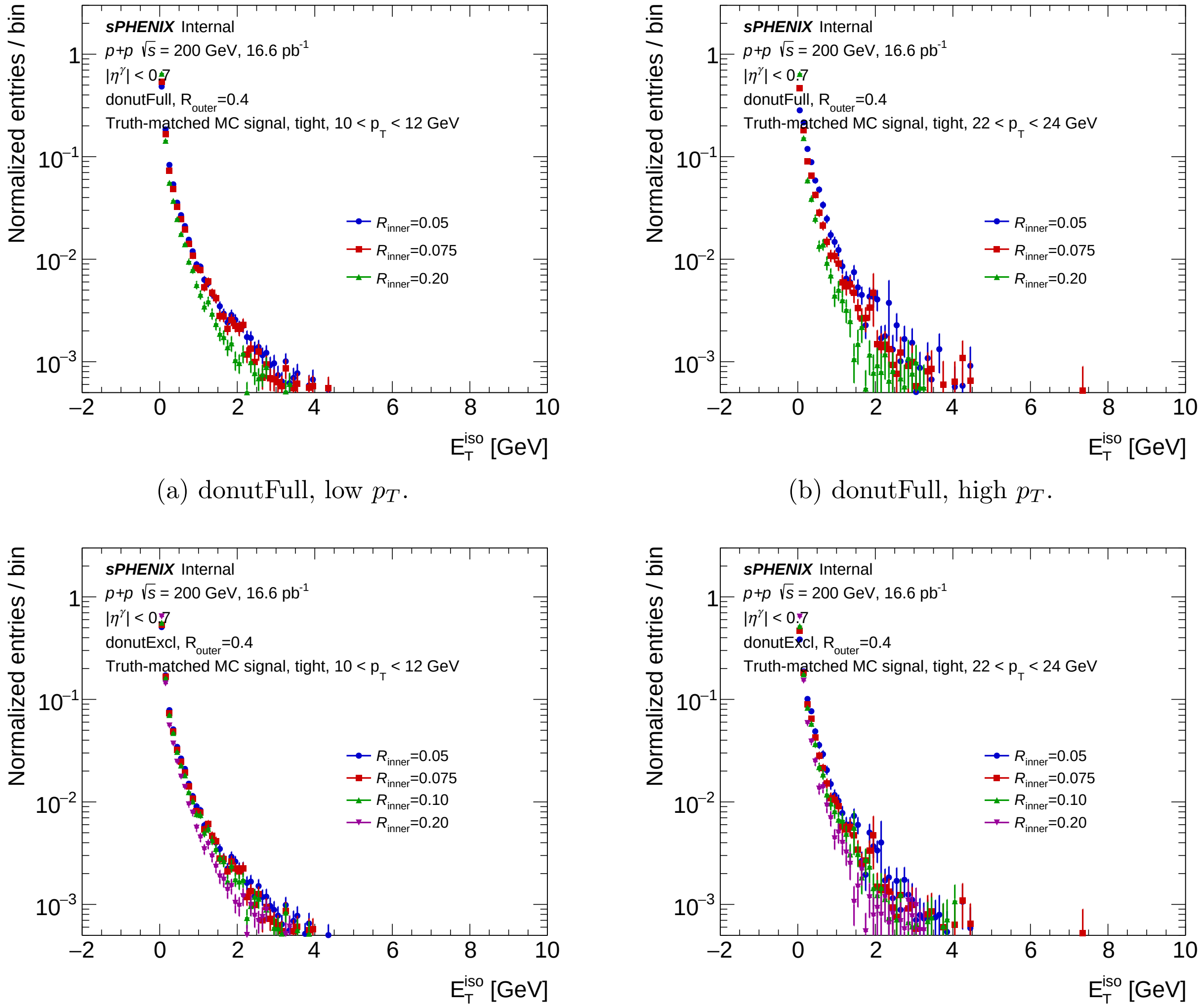
(c) donutExcl, low p_T.



(d) donutExcl, high p_T.

Figure 2: Tight-region isoET distributions in DATA for different donut inner R at two representative p_T bins. As R_{inner} grows, the annulus shrinks and the distribution narrows / shifts toward zero. At low p_T the distribution is broader; at high p_T it collapses toward the iso-cut edge.

Same in truth-matched MC signal (photon MC)



(a) donutFull, low p_T.

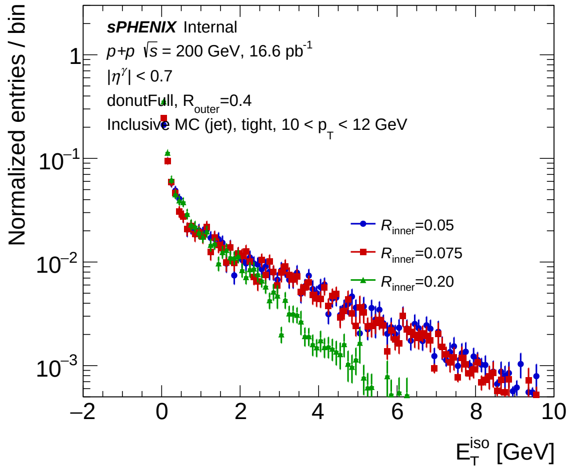
(b) donutFull, high p_T.

(c) donutExcl, low p_T.

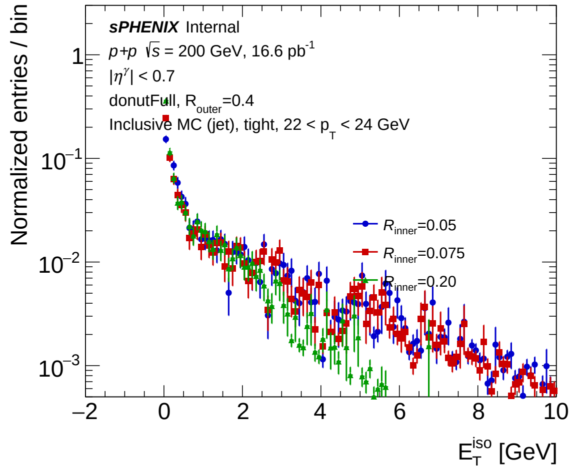
(d) donutExcl, high p_T.

Figure 3: Same as above for the truth-matched MC signal (merged photon MC, `MC_efficiency_bdt_<var>.root`). Distributions peak near zero and are narrower than data, as expected for isolated prompt photons. Shape differences across R_{inner} are subtler because the underlying-event pedestal is smaller in signal-only MC.

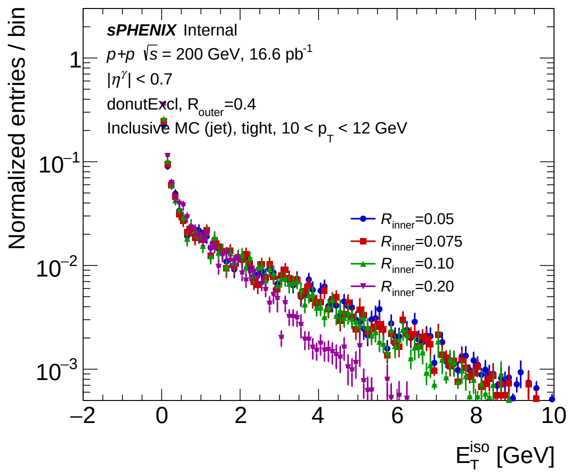
Same in inclusive MC (photon + jet, weighted)



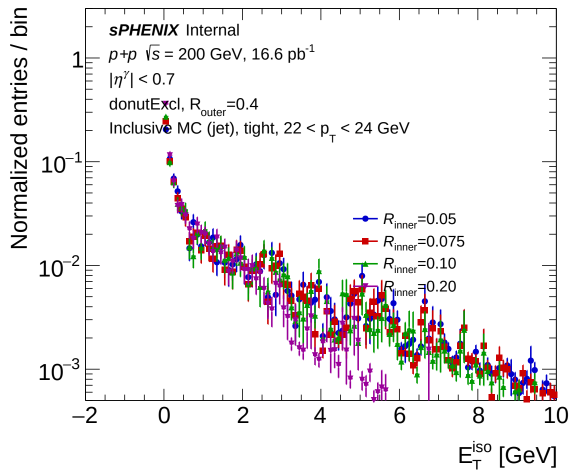
(a) donutFull, low p_T.



(b) donutFull, high p_T.



(c) donutExcl, low p_T.



(d) donutExcl, high p_T.

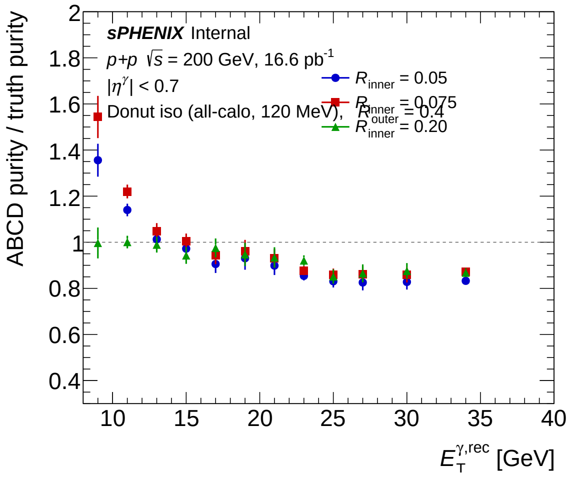
Figure 4: Same as above for inclusive MC (photon MC + jet MC, each weighted by its cross section via MergeSim). This is the direct MC analogue of the data plot above and should match data at cut1. The shape mirrors data closely, confirming the donut definition's behavior on the background-dominated tight-region population.

ABCD-on-MC closure

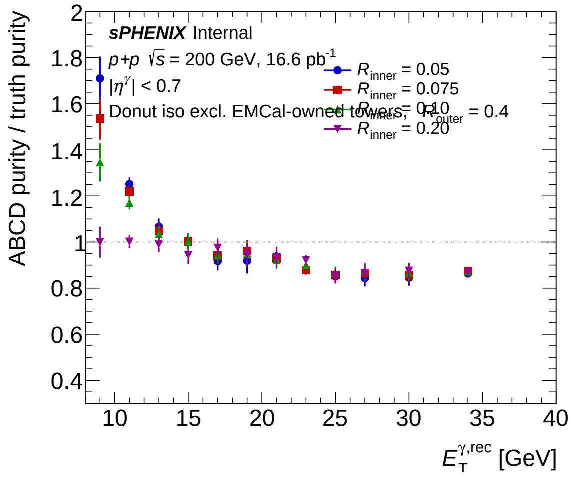
The key validation: does the ABCD background subtraction reproduce the truth purity on MC? The ratio $r = \text{purity}_{\text{ABCD}}/\text{purity}_{\text{truth}}$ is stored in `g_mc_purity_fit_ratio`; $r = 1$ is perfect closure.

Table 3: ABCD/truth closure ratio at selected p_T bins and mean absolute deviation from unity. Lower $\langle|1-r|\rangle$ means better closure.

Variant	$r(9)$	$r(15)$	$r(21)$	$r(28)$	$r(34)$	$\langle\|1-r\|\rangle$
nom (reference)	1.795	0.972	0.910	0.842	0.832	0.185
donutFull_005	1.356	0.972	0.899	0.826	0.833	0.136
donutFull_0075	1.543	1.004	0.931	0.861	0.873	0.138
donutFull_02	**0.997**	**0.943**	**0.934**	**0.865**	**0.868**	**0.069**
donutExcl_005	1.710	1.004	0.937	0.844	0.864	0.164
donutExcl_0075	1.536	1.002	0.929	0.866	0.874	0.137
donutExcl_01	1.346	1.003	0.922	0.877	0.872	0.113
donutExcl_02	**0.999**	**0.942**	**0.933**	**0.865**	**0.868**	**0.070**



(a) donutFull family.



(b) donutExcl family.

Figure 5: Closure of ABCD-derived purity against truth-matched purity on MC. The dashed line at 1.0 is perfect closure. All variants (including nominal) show a common low-p_T over-prediction and high-p_T under-prediction; inner $R = 0.20$ variants are closest to unity across the full p_T range.

Physics observations

- **Closure is the headline result.** The nominal topo04 iso has an 18.5% average deviation from unity; donut variants with $R_{\text{inner}} = 0.20$ reduce this to 6.9–7.0%. Other donut variants fall in between (11–16%).
- **Low-p_T non-closure is universal.** At $p_T = 9\,\text{GeV}$ the ratio is 1.35–1.80 across *all* variants including nominal. This is probably driven by the fit-region statistics at the edge of the reco p_T binning, not by the iso definition itself.
- **High-p_T under-prediction is also universal**, with $r \approx 0.83$–0.87 at $34\,\text{GeV}$ across all variants. ABCD consistently under-estimates truth purity at high p_T regardless of the iso definition.
- **Inner R dominates purity, not outer-cone treatment.** donutFull vs. donutExcl at matched R_{inner} give truth purities within ± 0.05 at low p_T and $\lesssim 0.02$ at high p_T. `donutFull_02` and `donutExcl_02` are numerically identical to four decimals because the fitted iso cut is the same (see derived (b, s)).

- **Tighter inner R gives higher truth purity at low p_T.** For the Excl family, $y(9\,\mathrm{GeV})$ monotonically decreases from 0.648 at $R_{\mathrm{inner}} = 0.05$ to 0.496 at $R_{\mathrm{inner}} = 0.20$. Interpretation: larger inner R admits less annulus area, giving a looser absolute iso cut that keeps more background.

Caveats and follow-ups

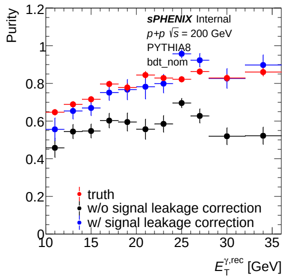
- **Outer cone includes HCal.** The donut is not pure EMCal. A pure-EMCal donut requires new inner-ring branches `cluster_iso_{005,0075,01,02}_emcal` at matching threshold (logged in `reports/anatreemaker_reprocess_TODO.md`).
- `donutFull_01` **skipped** pending upstream `cluster_iso_01` branch (all-calo, 120 MeV, $R < 0.1$).
- **Degraded mirrors** (`Cluster_rbr.C`, `EtaMigrationStudy.C`, `compare_energy_response.C`) silently use the full outer cone without inner subtraction for cases 4/5 since they do not read `iso_emcalinnerr`. Do not run donut variants through these diagnostic utilities expecting donut semantics.
- **MC iso scale/shift** set to 1.0/0.0 for all donut variants because the nominal 1.2/0.1 was tuned against full-cone topo04. Retuning would be needed if these ever became systematic variants.
- **The low-p_T closure deviation is present in nominal too.** This is a pre-existing effect of the ABCD method at the binning edge, independent of the iso definition. Worth understanding before any purity systematic is claimed.

Reviewer pass matrix

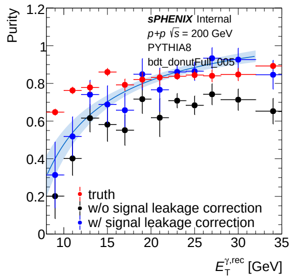
Artifact	Physics	Cosmetics	Re-derivation	Fix-valida
`use_topo_iso=4,5` switch	PASS (2 WARN util)	N/A	N/A	PASS
donut truth-purity scan PDFs	N/A	PASS (2 WARN)	PASS	N/A
donut closure PDFs	N/A	(not yet reviewed)	PASS	N/A
80 % iso cuts	N/A	N/A	PASS	PASS
`g_purity_truth` (7 variants)	N/A	N/A	PASS	PASS
`g_mc_purity_fit_ratio` (7 variants)	N/A	N/A	PASS	PASS

Per-variant MC purity closure

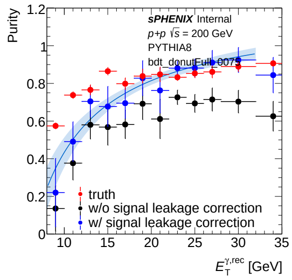
Each panel below shows, for one variant, the ABCD-derived purity on MC (points with error bars), the fit to the ABCD points (solid line), the signal-leakage-corrected ABCD purity, and the truth-matched purity (red markers). For a closure-good variant, the ABCD points/fit should track the truth markers; deviations mark where the ABCD subtraction biases the purity estimate at that p_T. Nominal topo04 iso (`bdt_nom`) is included as a reference.



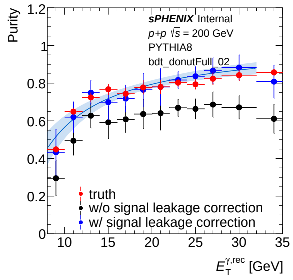
(a) nom (reference).



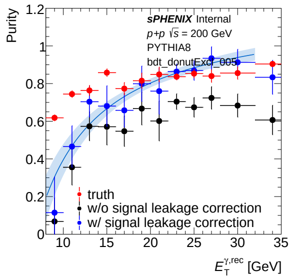
(b) donutFull_005.



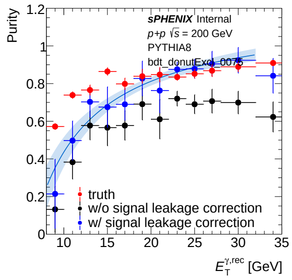
(c) donutFull_0075.



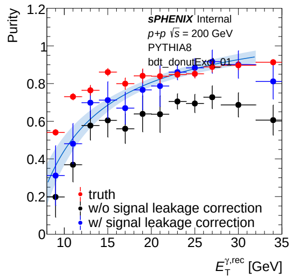
(d) donutFull_02.



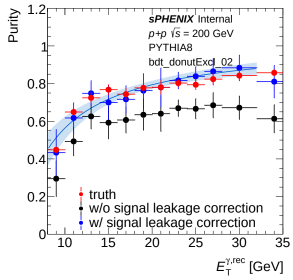
(e) donutExcl_005.



(f) donutExcl_0075.



(g) donutExcl_01.



(h) donutExcl_02.

Figure 6: Per-variant MC purity closure plots. Each panel: ABCD-derived purity points (from `gpurity`/`gpurity_leak`), the ABCD fit, and truth-matched purity (`g_purity_truth`). The quality of agreement between ABCD and truth across p_T is the closure check summarized in the previous section's ratio table. `donutFull_02` and `donutExcl_02` show the tightest truth–ABCD agreement, especially at low p_T.

Final cross section per variant

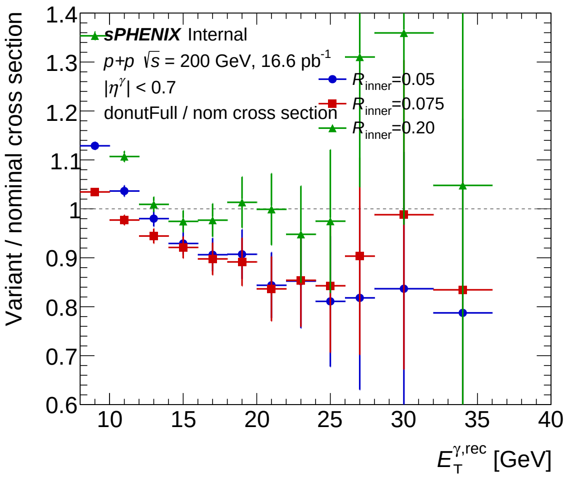
Donut / nominal ratio

The following ratio is defined per reco p_T bin on the unfolded signal yield `h_unfold_sub_result` (which is the background-subtracted, unfolded yield used to compute $d^2\sigma/d\eta\, dp_T$): $r = \text{yield}_{\text{donut}}/\text{yield}_{\text{nom}}$.

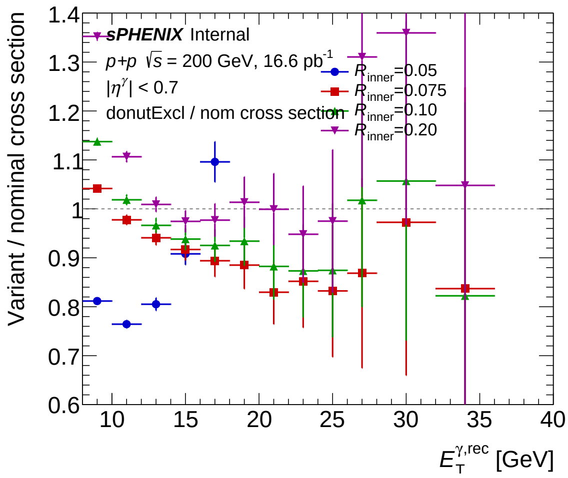
Table 4: Donut / nominal unfolded-yield ratio at selected p_T bins. Mean absolute deviation from unity averaged over the 12 reco p_T bins.

Variant	$r(9)$	$r(15)$	$r(21)$	$r(28)$	$r(34)$	$\langle\lvert 1-r\rvert\rangle$
donutFull_005	1.129	0.929	0.844	0.818	0.788	0.130
donutFull_0075	1.035	0.921	0.837	0.903	0.835	0.097
donutFull_02	1.354	0.974	0.999	1.310	1.048	0.138
donutExcl_005 (anomalous)	0.812	0.908	1.937	**−6.02**	**−0.04**	1.665
donutExcl_0075	1.042	0.917	0.830	0.869	0.837	0.104
donutExcl_01	1.137	0.938	0.882	1.018	0.822	0.096
donutExcl_02	1.352	0.974	0.999	1.310	1.048	0.137

donutExcl_005 anomaly. This variant yields negative unfolded signal at $p_T \geq 28\,\text{GeV}$, implying the ABCD background estimate exceeds the tight+iso count in those bins. Looking at the raw `h_unfold_sub_result` across p_T: values at $p_T = 22$–26 are anomalously large (2–6× nominal) before the bin-11 sign flip to −5.66 and bin-12 −0.81. Truth purity for the same variant is healthy (0.65–0.86), so the pathology is in yield extraction, not purity. Likely an interaction between the tight-iso count and the R-factor estimator at low statistics with the tightest inner R + ownership-excluded outer cone. Recommend: re-run with higher MC statistics (more condor jobs per sample) or drop this variant from the scan.



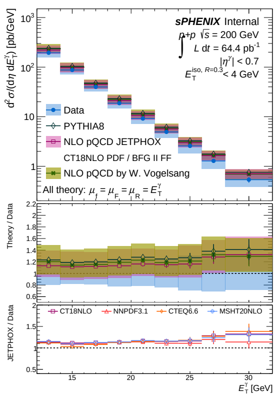
(a) donutFull / nominal.



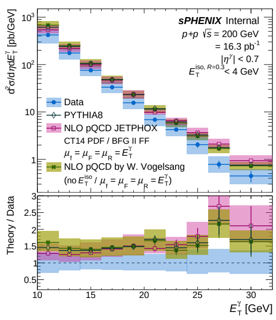
(b) donutExcl / nominal.

Figure 7: Unfolded-yield ratio (donut / nominal) vs. reco p_T. Dashed line at unity. Excluding `donutExcl_005`, variants agree with nominal to $\sim$ 10–14% on $\langle|1-r|\rangle$. The `R_inner`=0.20 variants diverge upward at low p_T (ratio up to 1.35) and stay within $\pm5\%$ of nominal at mid-p_T. `donutExcl_005` is plotted but pT $\geq 26\,\text{GeV}$ points fall off-scale (negative / $\gg 1.4$).

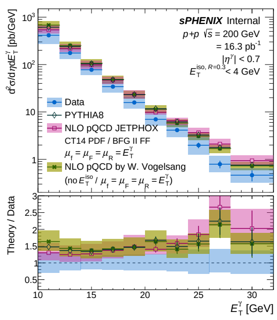
Per-variant final cross section



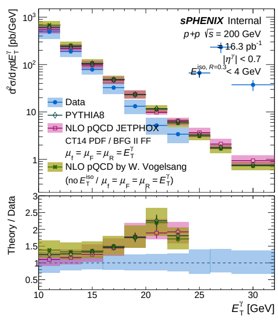
(a) nom.



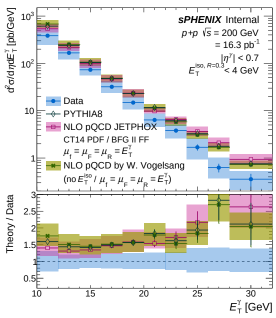
(b) donutFull_005.



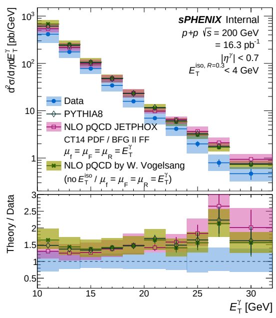
(c) donutFull_0075.



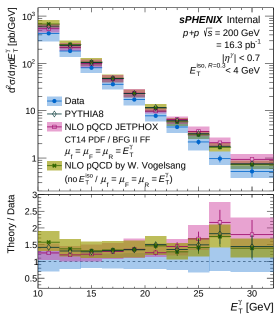
(d) donutFull_02.



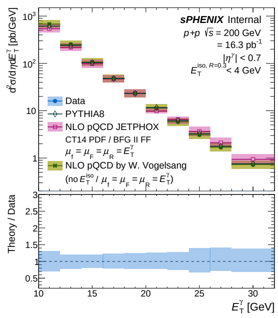
(e) donutExcl_005 (anomalous).



(f) donutExcl_0075.



(g) donutExcl_01.



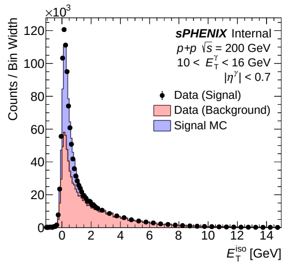
(h) donutExcl_02.

Figure 8: Per-variant final isolated-photon cross section $d^2\sigma/d\eta\,dp_T$ vs. p_T^γ, produced by `plot_final_selection.C`. Nominal topo04 iso (first panel) for reference. The `donutExcl_005` panel shows the high-p_T anomaly noted above.

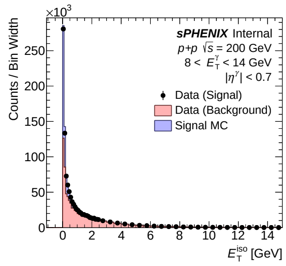
Isolation ET template fits (CONF)

Produced by `plotting/CONF_plots.C` from `data_histo_bdt_<var>.root` (tight and non-tight data) and `MC_efficiency_bdt_<var>.root` (MC signal template). Two p_T bin ranges shown as representatives: **8–14 GeV** (bins 0–2) and **20–26 GeV** (bins 6–8). The fit overlays the MC signal shape + a non-tight-derived background template onto the tight-region data, extracting the signal fraction per pT range. Full 4-range sets (0–2, 3–5, 6–8, 9–9) for all 7 donut variants are in `plotting/figures/h1D_iso_fit_bdt_donut*.pdf`; the selection report includes them in each per-config section via the standard PLOT_GROUPS "Isolation fit (CONF)" entry.

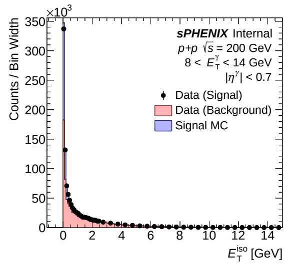
Low p_T bin range: $8 < p_T < 14\,\mathrm{GeV}$ (bins 0–2)



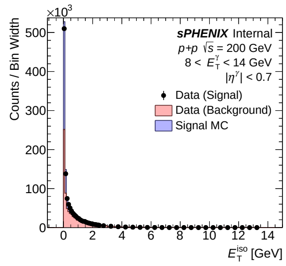
(a) nom.



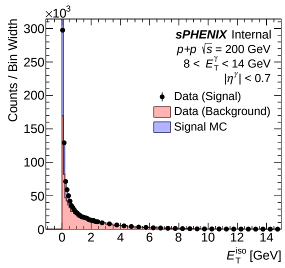
(b) donutFull_005.



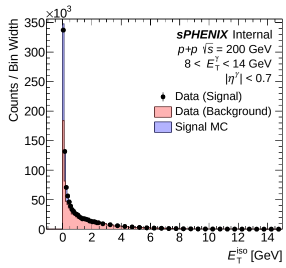
(c) donutFull_0075.



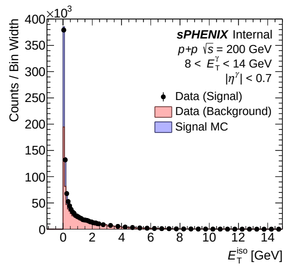
(d) donutFull_02.



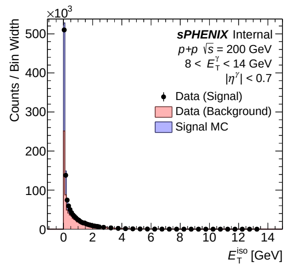
(e) donutExcl_005.



(f) donutExcl_0075.



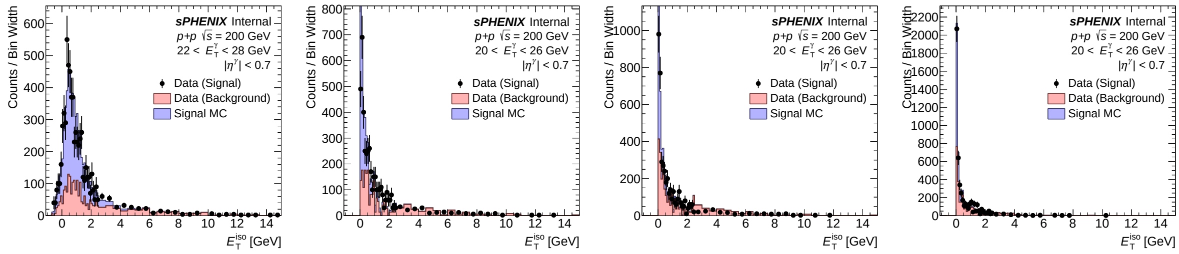
(g) donutExcl_01.



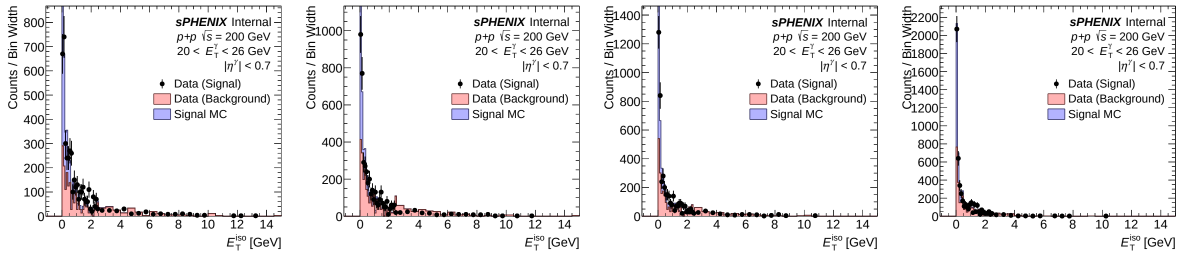
(h) donutExcl_02.

Figure 9: Isolation-ET templates vs. tight data (points), with MC signal template (blue fill) and non-tight data-derived background template (red fill), for $8 < p_T < 14\,\mathrm{GeV}$. Nominal reference and all 7 donut variants.

High p_T bin range: $20 < p_T < 26\,\mathrm{GeV}$ (bins 6–8)



(a) nom. (b) donutFull_005. (c) donutFull_0075. (d) donutFull_02.



(e) donutExcl_005. (f) donutExcl_0075. (g) donutExcl_01. (h) donutExcl_02.

Figure 10: Same as above for $20 < p_T < 26\,\mathrm{GeV}$. The `donutExcl_005` high-$p_T$ pathology noted in the cross-section section shows up here as the signal+background templates failing to cover the tight-region data shape.

BDT model alternatives: `base_v0` and `base_v0E`

Cross-check on BDT model choice: two additional variants apply earlier model versions (`base_v0`, `base_v0E`) across both ET bins via `bdt_et_bin_models=["base_v0","base_v0"]` (resp. v0E). Nominal topo04 iso and nominal tight/non-tight BDT cuts are kept. Both $cluster\_bdt\_CLUSTERINFO\_CEM$ and $*\_base\_v0E$ branches already exist in the slimtree; no re-apply needed. HTCondor cluster 2734512 (2 jobs, ∼20 min each).

Summary numbers

Table 5: Truth purity, ABCD/truth closure, and unfolded-yield ratio to nominal, at selected p_T bins. Lower $\langle|1-r|\rangle$ is better closure. Higher $|1-r_{\mathrm{xsec}}|$ means larger deviation of the cross section from nominal.

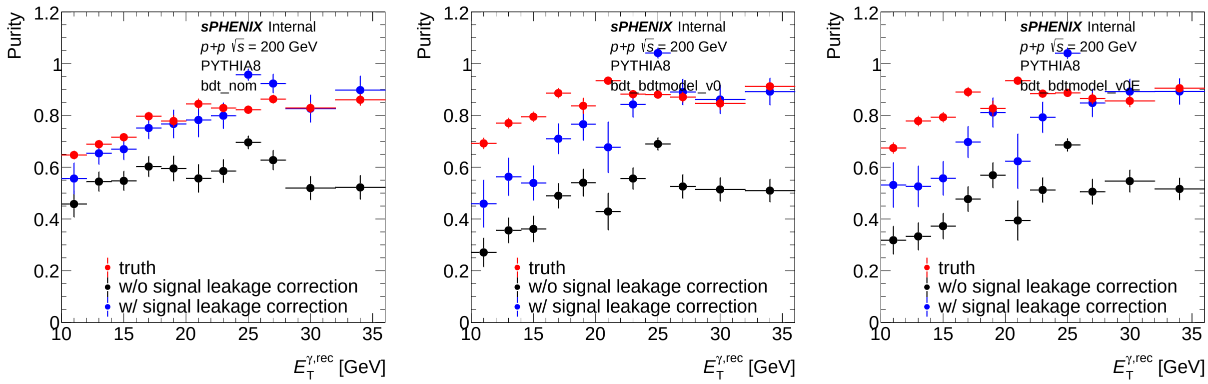
	Truth purity			ABCD/truth closure		
Variant	$y(9)$	$y(21)$	$y(34)$	$r(9)$	$r(21)$	$\langle\|1-r\|\rangle$
nom (reference)	0.581	0.743	0.797	1.795	0.910	0.185
bdtmodel_v0	0.544	0.744	0.782	2.177	0.973	0.226
bdtmodel_v0E	0.557	0.753	0.786	2.491	0.961	0.265

Table 6: Unfolded-yield ratio (variant / nominal) across the p_T spectrum.

Variant	$r(9)$	$r(15)$	$r(21)$	$r(28)$	$r(34)$	$\langle\|1-r\|\rangle$
bdtmodel_v0	0.751	0.980	1.014	1.183	1.604	0.196
bdtmodel_v0E	0.760	0.974	1.006	1.219	1.742	0.223

Observations. (1) Truth purity for the v0/v0E models is close to (but slightly lower than) nominal at low p_T (0.54–0.56 vs. 0.58), and essentially identical at high p_T. (2) **ABCD closure is worse** for both alternatives: $\langle|1-r|\rangle = 0.226$ (v0) and 0.265 (v0E) vs. 0.185 nominal. The low-p_T overprediction is particularly strong ($r = 2.18$ and 2.49 at $p_T = 9\,\mathrm{GeV}$ vs. 1.80 nominal), suggesting the ABCD R-factor assumption is less satisfied with these BDT discriminants. (3) Cross section at high p_T diverges sharply from nominal: 1.60× and 1.74× at $p_T = 34\,\mathrm{GeV}$ respectively. The v0/v0E models either admit substantially more tight+iso yield (looser BDT) or have larger non-closure absorbing into the subtracted "signal" yield. Given that the cross-section ratio grows with p_T while closure degrades at low p_T, the high-p_T excess is likely driven by poorer signal–background separation of the older BDT variants in the region where statistics are thin.

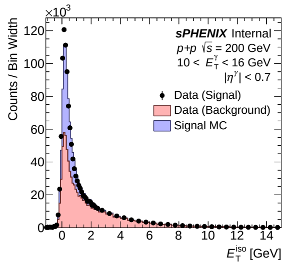
Per-variant MC purity closure



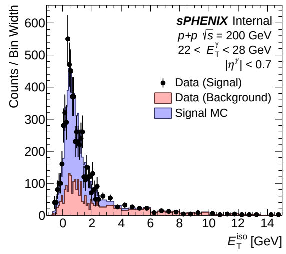
(a) nom (reference). (b) bdtmodel_v0. (c) bdtmodel_v0E.

Figure 11: MC purity closure for the two alternative BDT models. Each panel shows the ABCD-derived purity (points), the fit, and the truth-matched purity (red markers).

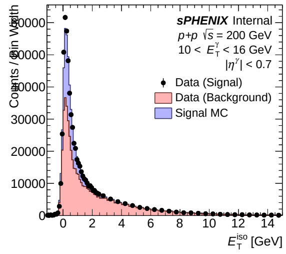
Isolation ET template fits



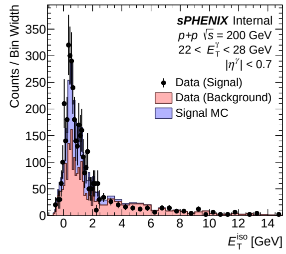
(a) nom, $8 < p_T < 14$.



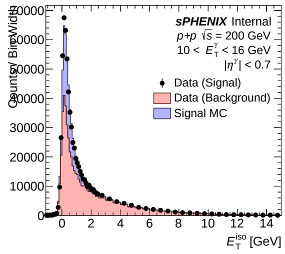
(b) nom, $20 < p_T < 26$.



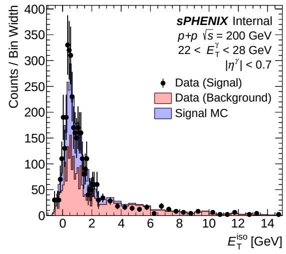
(c) v0, $8 < p_T < 14$.



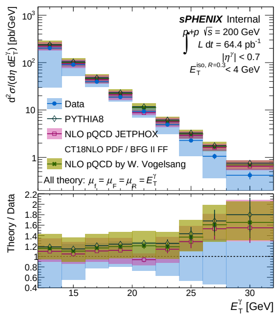
(d) v0, $20 < p_T < 26$.



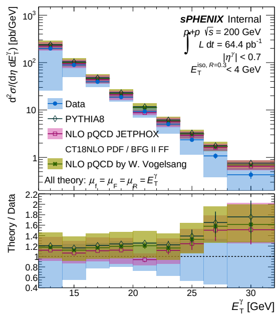
(e) v0E, $8 < p_T < 14$.



(f) v0E, $20 < p_T < 26$.



(g) Final $d^2\sigma$, v0.



(h) Final $d^2\sigma$, v0E.

Figure 12: CONF template overlays (`h1D_iso_*`: tight data points, MC signal template in blue fill, non-tight background template in red fill) for nominal and the two alternative BDT models at two p_T ranges, followed by the unfolded final cross section per alternative model. Poorer template agreement for v0/v0E at high p_T is consistent with the weaker closure reported above.

Final cross section with MC purity correction turned on

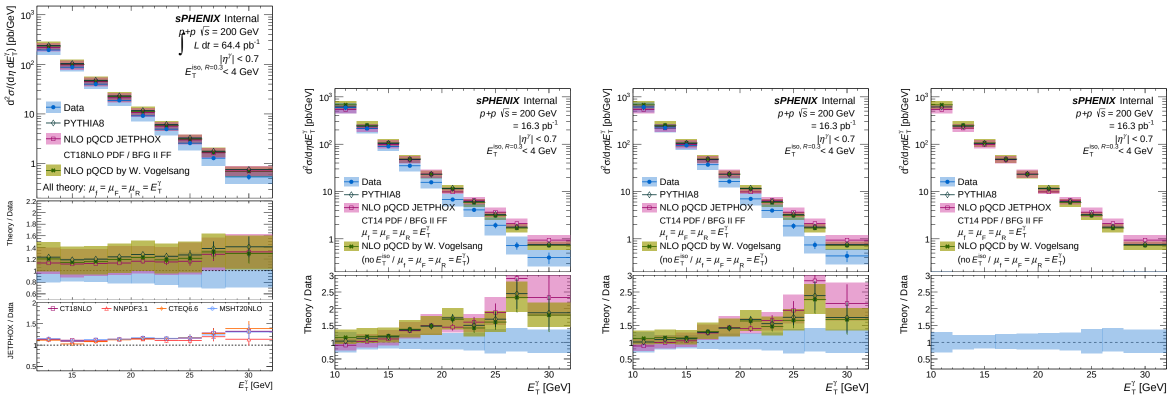
Same 7 donut variants as Figure 8, now run with `mc_purity_correction=1` in the config (the existing `config_bdt_mc_purity_correction.yaml` mechanism applied per-variant). In this mode `CalculatePhotonYield.C` uses the MC-as-data truth purity (from `Photon_final_bdt_<var>_mcpc_mc.root`) as a multiplicative correction when extracting the data purity, rather than trusting the ABCD fit alone. Truth purity and ABCD-on-MC closure are unchanged from the non-mcpc run (they are MC-only quantities); the difference is in the data yield.

Table 7: mcpc: cross-section ratio (variant / nominal, both without mcpc for the nominal denominator) and mean absolute deviation. Compare with the corresponding non-mcpc row from the earlier cross-section table.

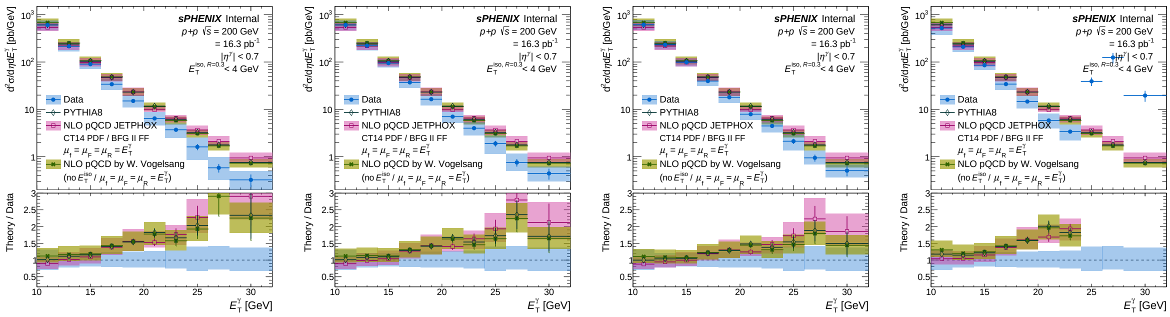
Variant	$r_{xs}(9)$	$r_{xs}(21)$	$r_{xs}(34)$	$\langle\lvert 1 - r_{xs}\rvert\rangle$
donutFull_005_mcpc	1.534	0.844	0.655	0.282
donutFull_0075_mcpc	1.599	0.837	0.725	0.257
donutFull_02_mcpc	1.352	0.999	0.910	0.129
donutExcl_005_mcpc	1.619	0.937	0.673	0.321
donutExcl_0075_mcpc	1.607	0.929	0.726	0.264
donutExcl_01_mcpc	1.540	0.922	0.714	0.220
donutExcl_02_mcpc	1.350	0.999	0.911	0.129

Note on `donutExcl_005`. Without MC purity correction this variant showed negative unfolded yields at $p_T \geq 28$ GeV ($\langle\lvert 1 - r\rvert\rangle = 1.665$). With mcpc on, the yields are all positive and $\langle\lvert 1 - r\rvert\rangle$

drops to 0.321. MC purity correction effectively rescues the pathological variant by using the MC truth purity in place of the badly-behaved ABCD purity at high p_T.



(a) nom (no mcpc, reference). (b) donutFull_005_mcpc. (c) donutFull_0075_mcpc. (d) donutFull_02_mcpc.



(e) donutExcl_005_mcpc. (f) donutExcl_0075_mcpc. (g) donutExcl_01_mcpc. (h) donutExcl_02_mcpc.

Figure 13: Per-variant final cross section with MC-purity correction enabled (`mc_purity_correction=1`). Nominal (no mcpc) panel included as reference. The `donutExcl_005_mcpc` panel no longer shows the negative-yield pathology at high p_T.

Full cone, no inner veto ring

Two additional cross-check variants use the same tower-based isolation (R=0.4 outer, 120 MeV threshold, all-calo or ownership-excluded EMCal) but *skip the inner-R subtraction*. Implemented by setting `iso_emcalinnerr=0.0` in the `use_topo_iso=4/5` block, so the isolation value is the raw full cone $E_T^{\rm iso}$ = `cluster_iso_04` (or `_excl_04`) with no annulus subtraction. 80 % iso cuts rederived via the two-pass workflow:

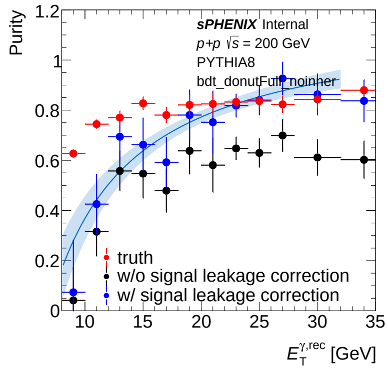
Table 8: Derived 80 % iso cuts for the full-cone (no-inner-ring) variants.

Variant	b [GeV]	s
donutFull_noinner	0.0712	0.01655
donutExcl_noinner	0.2069	0.01889

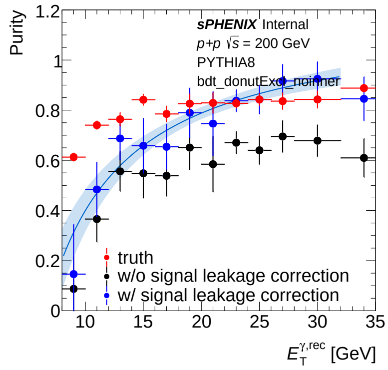
Table 9: Truth purity, closure, and cross-section ratio vs. nominal for the full-cone variants.

	Truth purity			Closure / xsec	
Variant	$y(9)$	$y(21)$	$y(34)$	$\langle \lvert 1 - r_{\text{cls}} \rvert \rangle$	$\langle \lvert 1 - r_{\text{xs}} \rvert \rangle$
nom (reference)	0.581	0.743	0.797	0.185	0
donutFull_noinner	0.653	0.765	0.827	0.197	0.236
donutExcl_noinner	0.632	0.780	0.843	0.153	0.167

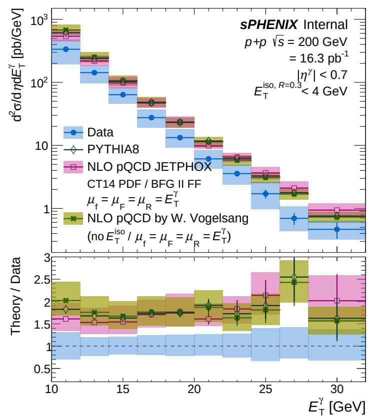
Observations. Full-cone iso without inner-ring subtraction gives purity *higher* than nominal at all p_T (by $\sim$ 0.05–0.07 at low p_T), but closure is only marginally better than nominal for `donutExcl_noinner` ($\langle \lvert 1-r \rvert \rangle = 0.153$ vs. nominal 0.185) and slightly *worse* for `donutFull_noinner` (0.197). Cross-section ratios at high p_T are closer to unity (0.78–0.89) than some of the donut variants, but mean deviation is larger than the best donut ($R_{\text{inner}} = 0.20$) variants.



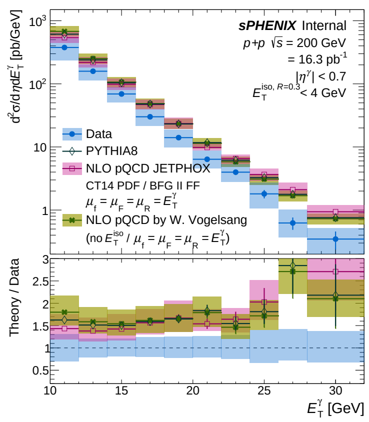
(a) donutFull_noinner purity.



(b) donutExcl_noinner purity.



(c) donutFull_noinner final.



(d) donutExcl_noinner final.

Figure 14: Full-cone (no inner-ring) variants: per-variant MC purity closure (top row) and final $d^2\sigma/d\eta\, dp_T$ (bottom row).